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SYSTEMATIC REVIEW ARTICLE

Herbal Medicine as an Alternative Treatment in Autism Spectrum Disorder: A Systematic Review

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Abstract: Background: This review focuses on the efficacy and safety of herbal medicines in the management of autism spectrum disorder (ASD) in humans and animals.

Methods: PubMed, Scopus, Google Scholar, Web of Science, and Science Direct databases were searched up to October 30, 2016. The key terms used were "ASD", "Asperger", "autism", "healing plants", "herbal medicine", and "medicinal plants". In each database, the searches consisted of each of three key terms describing the disorder and subtypes plus each of the terms describing the therapy. All human and animal studies on the effects of herbs with the key outcome of change in autism symptoms were included. In vitro studies were excluded.

Results: From the publications perused in the initial database, 3157 results were identified, reviewed and a total of 23 studies were included. Preclinical studies using critically validated models were conducted, with some promising preliminary results. Data availability on controlled clinical studies is currently very limited. The use of different methodologies and the very small number of patients raise doubts about the effects of these preparations.

Conclusion: Available data do not yet allow us to suggest the effectiveness of herbal medicines as an add-on in the treatment of ASD symptoms.

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1. INTRODUCTION

Autism spectrum disorder (ASD) encompasses different syndromes characterized by neurodevelopmental disorders with some fundamental characteristics, which can manifest together or separately. These include, among others, communication difficulty by failure in the field of language, inability to use the imagination to deal with symbolic games, difficulty in socializing, restrictive and repetitive behavioral patterns [1].

In recent years, we have seen a significant increase in ASD diagnosis, evidenced as a "new epidemic." This could partly be explained due to deeper understanding of the disease and a significant improvement in diagnostic techniques [2].

Although the matter was much in evidence in the scientific community, there are still many controversies on the causes of this disease. In fact, a set of genetic and environmental factors appear to be involved, however, the true origins of ASD remain unclear [3]. Research shows that, currently, one in every hundred children is a spectrum carrier, which affects more boys than girls. In general, the disorder is installed in the first three years of life when the neurons that coordinate communication and social relationships fail to make the necessary connections [4].

Various pharmacological and non-pharmacological treatments have been proposed to combat ASD. In general, current pharmacological treatments, especially with antipsychotic drugs, are intended

for controlling neurological and behavioral symptoms such as irritability, insomnia and aggression. If, on one hand, these drugs adequately control these symptoms, side effects often lead to discontinuation [5].

Currently, the first-line treatments for ASD are restricted to non-pharmacological measures, especially behavioral therapies, such as TEACCH (Treatment and Education of Autistic and Related Communication Handicapped Children) [6] and/or ABA (Applied Behavior Analysis) [7]. In fact, these interventions have shown promising results in reducing the symptoms, mainly in cases of difficulty in communicating and socializing [8].

The current context of ASD, particularly regarding the increase of prevalence and the absence of effective pharmacological treatment, generated a miscellany of complementary and alternative medicine (CAM), promising satisfactory results [9]. In this sense, research indicates that each family with an ASD child has tried an average of 7 treatments, including many CAMs. In fact, it has recently been estimated that 28% of children with ASD have been treated with CAMs [10, 11].

Several CAMs have been employed in the treatment of ASD, especially the elimination or addition of special diets [12]. Moreover, although widely used throughout the world [13], the use of herbal medicine as a regular therapy for ASD treatment is still being widely investigated and/or registered. Therefore, the aim of this systematic review is to give an overview of the use of herbal medicine in ASD.

2. METHODS

A systematic review was performed based on articles published from October 1976 to October 2016. Medline (by Pubmed), Science

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Direct, Web of Science (by ISI Web of Knowledge) and Scopus databases were used in order to identify the most appropriate studies. The key terms used (MeSH database) were “ASD”, “Asperger”, “autism”, “healing plants”, “herbal medicine”, and “medicinal plants”. In each database, the searches included each of the three key terms describing the disorder and subtypes plus each of the terms describing the therapy.

In order to be considered in this analysis, articles had to meet the following two inclusion criteria: the studies need to be performed with crude extract, mixtures of herbs or isolated fractions from herbal extracts and not with isolated compounds or mixed formulations; the articles would have to be published in English, Spanish or Portuguese. To ensure the consistency and reproducibility of the process, we followed the recommendations of the Cochrane Collaboration and PRISMA statement [14, 15] for systematic reviews. Two independent reviewers evaluated the articles by their titles and abstracts, selecting them according to the inclusion/exclusion criteria set out above. Subsequently, the articles selected in this initial screening were carefully evaluated and excluded or permanently maintained in the review.

3. RESULTS AND DISCUSSION

Our literature search identified 3157 studies. After the title/abstract screening, 112 publications were thoroughly evaluated. After detailed evaluation, 20 studies were included; through hand search of the literature, three additional studies were included, making a total of 23. Detailed information on collected data and the medicinal species presented in the selected studies are shown in Table 1. In general, few preclinical and clinical studies have been critically conducted and published in the selected databases. In most cases, review articles that addressed generic aspects of the alternative therapies used to control the symptoms of ASD were found.

When we look closely at the preclinical studies, we realize that there is no fully adequate model that mimics all behavioral changes present in ASD. In addition, most of the preclinical studies were conducted over the last decade [16-21]. In fact, of the six studies found, five used prenatal administration of valproic acid as a model for the induction of some symptoms present in the ASD, including increased levels of anxiety and locomotor hyperactivity. In this case, the administration of different preparations obtained from *Camellia sinensis* (L.) Kuntze [16], *Panax ginseng* C.A. Mey. [18, 20], *Bacopa monnieri* L. [17] and *Oryza sativa* L. plus *Bombyx mori* L. [21] were able to improve the symptoms of ASD in rodents, showing relevant anxiolytic, antioxidant and neuroprotective effects.

In a similar way to the above-mentioned data, a recent study conducted by Muneoka *et al.* [19] used an experimental model based on the prenatal administration of 5-bromo-2'-deoxyuridine in rats (BrdU-rats). Currently, it is known that this experimental model induces hyperlocomotor activity in rat offspring, converting to a useful animal model for the study of attention-deficit hyperactivity disorder (ADHD) and ASD. In this study, researchers used a preparation called Yokukansan (YKS), which consists of a mixture of seven kinds of dried medicinal herbs: *Atractylodis lanceae rhizoma*, *Poria* mushroom, *Cnidii rhizoma*, *Angelicae radix*, *Bupleuri radix*, *Glycyrrhizae radix* and *Uncariae ramulus et uncus* (Uncaria stem with hooks). The administration of YYS was able to positively influence the cerebellar serotonin metabolism of Sprague-Dawley rats, leading to a substantial improvement in cognitive ability and reducing locomotor agitation.

Similar to studies using animal models, few controlled clinical studies in ASD patients were conducted. Four of them [22-25] used only one medicinal species, while two others [26, 27] used traditional herbal mixtures.

In a preliminary open trial [24], three male patients with ASD, diagnosed by ICD-10 criteria, were treated with an extract obtained from *Hypericum perforatum* L. (St John's Wort; 20 mg/day) for 4 weeks. Subjects were included in the study if their eye contact and expressive language were inadequate for their developmental level and if they had not tolerated or responded to other psychopharmacologic treatments. According to this study, St John's Wort was modestly effective in short-term treatment only regarding irritability in some patients.

In another observational study [23], three patients received 100 mg of *Ginkgo Biloba* EGb 761 (twice a day), for 4 weeks, and showed very little improvement in behavioral symptoms. Moreover, Hasanzadeh *et al.* [22] decided to assess the effectiveness of *Ginkgo biloba* extract (Ginko T.D., Tolidaru, Iran) as an adjunctive agent to risperidone in the treatment of ASD. Forty-seven outpatients with a diagnosis of ASD according to DSM-IV-TR criteria, ages between 4 and 12 years old, were assigned to this double-blinded clinical trial and were randomly divided into two groups. One group received risperidone plus Ginko T.D. and the other received risperidone plus placebo. The risperidone dose was 1-3 mg/day; for Ginko T.D., the dose was 80 mg/day for patients under 30 kg and 120 mg/day for patients above 30 kg. The results show that the association of risperidone with *Ginkgo biloba* extract did not lead to improvements of the ASD symptoms evaluated. Therefore, its effect as an add-on therapy remains completely open and requires further investigation.

Niederhofer [25] evaluated three male outpatients (age range 18.4-22.2 years old; mean = 21.3 years; SD = 4.1 years), meeting ICD-10 criteria for ASD, treated with *Panax ginseng* (250 mg daily) for 4 weeks. Patients were free of medication for at least 4 weeks before the beginning of the study. During that time, there were no changes regarding the behavioral symptoms in patients. Since *Panax ginseng* did not cause any significant improvement, its effect as an add-on therapy also remains open and requires further studies.

Unlike previous reports regarding the use of an isolated medicinal species, two studies with herbal mixtures have shown promising results. In one, Wake *et al.* [27] conducted a 12-week prospective open-label survey of YYS in 20 children and adolescents, age 6 to 17 years, diagnosed with ASD. The children and adolescents received YYS in the dosage range of 2.5-7.5 g/day for 12 weeks. The preliminary data suggest that YYS may be effective and well tolerated for the treatment of severe irritability/agitation and hyperactivity/noncompliance in children and adolescents subjected to this study. However, given the characteristics of this trial, the present findings should be taken cautiously, and large-scale placebo-controlled studies are needed in order to elucidate the efficacy and tolerability of this preparation.

In another study conducted by Chan *et al.* [26], thirty children with ASD were evenly assigned to receive a daily intranasal administration of the Chinese patented herbal formula (patent number ZL 2008 1 0176088.7) or no treatment. Their executive functions, behavioral problems and electroencephalographic activity during an executive control task were evaluated before and after six months of treatment. After treatment, the experimental group showed significantly improved inhibitory control, mental flexibility, planning, as well as daily reduction of dysexecutive behaviors. In contrast, the control group showed no significant changes in executive functions or neural system activity.

Although some clinical studies have shown encouraging results, there are important limitations that raise doubts regarding the actual effectiveness of these preparations. The small number of individuals and duration of studies may represent inadequate variables to determine the complete effect of these preparations. Moreover, the environment in which the clinical evaluations were performed was not ideal to be considered as a sensitive indicator during the test.

Table 1. Family and medicinal species cited in preclinical, clinical or review studies involving autism spectrum disorder.

Family	Scientific name	Study	Country	References
Acanthaceae	<i>Adhatoda vasica</i> L.	R	Saudi Arabia	[35]
Amaryllidaceae	<i>Allium sativum</i> L.	R	Canada	[34]
Apiaceae	<i>Centella asiatica</i> (L.) Urban	R	Iran	[29; 32]
Araceae	<i>Acorus calamus</i> L.	R	Iran, Saudi Arabia	[29; 32; 35]
	<i>Acorus tatarinowii</i> Schott	R	USA	[28]
Asparagaceae	<i>Ophiopogon japonicus</i> (L.f.)	R	USA	[28]
Asteraceae	<i>Artemisia absinthium</i> L.	R	Romania	[36]
	<i>Artemisia afra</i> Jacq.	R	Romania	[36]
	<i>Carthamus tinctorius</i> L.	R	Romania	[36]
	<i>Centaurea benedicta</i> L.	R	Romania	[36]
	<i>Crassocephalum bauchiense</i> (Hutch.)	R	Romania	[36]
	<i>Echinacea purpurea</i> (L.) Moench.	R	Canada	[34]
	<i>Galanthus woronowii</i> Losinsk.	R	Romania	[36]
	<i>Helichrysum cymosum</i> (L.) Less.	R	Romania	[36]
	<i>Lactuca scariola</i> L.	R	Romania	[36]
	<i>Leuzea carthamoides</i> (Willd.) DC.	R	Romania	[36]
	<i>Papaver somniferum</i> L.	R	Romania	[36]
	<i>Prunus amygdalus</i> Batsch.	R	Romania	[36]
	<i>Sesamum indicum</i> L.	R	Romania	[36]
	<i>Silybum marianum</i> L.	R	Romania	[36]
	<i>Sonchus eruca</i> L.	R	Saudi Arabia	[35]
	<i>Trichogyne repens</i> L.	R	Romania	[36]
	<i>Tussilago farfara</i> L.	R	USA	[28]
	<i>Veronia oligocephala</i> (DC.)	R	Romania	[36]
	<i>Xanthium strumarium</i> L.	R	Romania	[36]
Campanulaceae	<i>Lobelia inflata</i> L.	R	Iran	[32]
Cannabaceae	<i>Cannabis sativa</i> L.	R	EUA	[33]
Combretaceae	<i>Anogeissus leiocarpus</i> (DC)	R	Saudi Arabia	[35]
	<i>Terminalia arjuna</i> Wight & Arn.	R	Saudi Arabia	[35]
	<i>Terminalia chebula</i> Retz.	R	Saudi Arabia	[35]
Euphorbiaceae	<i>Emblica officinalis</i> Gaertn.	R	Saudi Arabia	[35]
Fabaceae	<i>Abrus precatorius</i> L.	R	Saudi Arabia	[35]
	<i>Astragalus membranaceus</i> Moench.	R	Iran	[29; 32]
	<i>Astragalus onobrychis</i> L.	R	USA	[28]
	<i>Medicago sativa</i> L.	R	Saudi Arabia	[34]
Ginkgoaceae	<i>Ginkgo biloba</i> L.	C, R	Iran, Italy, USA	[22; 23; 29; 31; 32]

Table (1) contd....

Family	Scientific name	Study	Country	References
Hypericaceae	<i>Hypericum perforatum</i> L.	C	Italy	[24]
Lamiaceae	<i>Mentha spicata</i> L.	R	Saudi Arabia	[34]
	<i>Salvia miltiorrhizae</i> Bunge	R	Saudi Arabia; USA	[28; 35]
	<i>Salvia officinalis</i> L.	R	USA	[30]
	<i>Salvia triloba</i> L.	R	Saudi Arabia	[35]
Lauraceae	<i>Cinnamomum verum</i> J.S. Presl.	R	USA	[28]
Liliaceae	<i>Asparagus cochinchinensis</i> (Lour.) Merr.	R	USA	[28]
Meliaceae	<i>Entandrophragma angolense</i> C. DC.	R	Saudi Arabia	[35]
	<i>Khaya senegalensis</i> (Desr.) A. Juss.	R	Saudi Arabia	[35]
Moraceae	<i>Morus alba</i> L.	R	USA	[28]
Myrtaceae	<i>Myrtus communis</i> L.	R	Saudi Arabia	[35]
Orobanchaceae	<i>Rehmannia glutinosa</i> (Gaertn.) Steud.	R	USA	[28]
Paeoniaceae	<i>Paeonia lactiflora</i> Pall.	R	Iran	[32]
Pinaceae	<i>Pinus pinaster</i> Aiton	R	USA	[30]
Piperaceae	<i>Piper nigrum</i> L.	R	Saudi Arabia	[35]
Poaceae	<i>Oryza sativa</i> L.	PC	India	[21]
Polygalaceae	<i>Polygala tenuifolia</i> Wild.	R	USA	[28]
Ranunculaceae	<i>Actaea racemosa</i> L.	R	Iran	[32]
	<i>Coptis chinensis</i> Franch.	R	USA	[28]
Rhamnaceae	<i>Ziziphus jujuba</i> Mill.	R	USA	[28]
Rosaceae	<i>Fragaria ananassa</i> Duch.	R	Canada	[34]
Rubiaceae	<i>Pavetta crassipes</i> K. Schum.	R	Saudi Arabia	[35]
Schisandraceae	<i>Schisandrae chinensis</i> (Turcz.) Baill	R	USA	[28]
Scrophulariaceae	<i>Bacopa monnieri</i> L.	PC	India	[21]
Solanaceae	<i>Withania somnifera</i> (L.) Dunal	R	Saudi Arabia	[35]
Theaceae	<i>Camellia sinensis</i> (L.) Kuntze.	PC	India	[16]
Umbelliferae	<i>Saposhnikovia divaricata</i> (Turcz.)	R	USA	[28]
	<i>Angelicae sinensis</i> (Oliv.) Diels	R	USA	[28]
Zingiberaceae	<i>Zingiber officinale</i> (Willd.)	R	Iran	[29; 32]
	<i>Zingiber officinale</i> Roscoe	R	USA	[28]
Zygophyllaceae	<i>Tribulus terrestris</i> L.	R	Saudi Arabia	[35]

C: Clinical studies. PC: Preclinical studies. R: Reviews studies.

Furthermore, evaluations carried out by mentors and parents are of inestimable value in the assessment of patients with autistic disorder during psychopharmacological treatment. Therefore, it is still premature to state that the use of phytotherapy can effectively provide concrete clinical benefits to patients with ASD.

In addition to the data referring to controlled preclinical and clinical studies, many review articles, not restricted to the topic, describe several medicinal species that are employed empirically in

the treatment of ASD symptoms. A total of 64 medicinal species were cited in these works, which, however, lack information on the clinical effectiveness of these compounds. Indeed, if we consider the extensive use of medicinal plants worldwide, it is likely that many other species are commonly employed for the control of ASD symptoms. However, the widespread use does not guarantee the effectiveness of these preparations. Ethnobotanical studies can identify these species and controlled preclinical trials can lead to

validation of its efficacy and safety. Thus, there is a real prospect that, after standardization of these preparations and mapping the major secondary metabolites, appropriate clinical trials can definitively prove or refute the effectiveness of these agents as an add-on treatment of ASD symptoms.

CONCLUSION

Although the use of herbal medicines is an ancient practice, preclinical and clinical studies have only focused on the effects of these preparations in ASD patients in the last decade. Few preclinical studies using critically validated models were conducted, although some preliminary results are promising. Data availability on controlled clinical studies is currently very limited. The use of different methodologies and the very small number of patients raise doubts regarding the effects of these preparations. Only the development of adequate preclinical models and the conduct of controlled clinical trials with scientifically validated methodology may be able to determine whether the use of different herbal medicines may be a viable alternative to conventional pharmacological treatments used to control ASD's symptoms.

LIST OF ABBREVIATIONS

ASD	=	Autism spectrum disorder
ADHS	=	Attention deficit hyperactivity disorder
CAM	=	Complementary and alternative medicine
YKS	=	Yokukansan

CONSENT FOR PUBLICATION

Not applicable.

CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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